Amjad Hmoud

Master’s Student in Data Science

About Me

Data Science master’s student with practical experience in building machine learning models and analyzing real-world datasets. Skilled in Python, data analysis, and predictive modeling, with a focus on delivering actionable insights.

Worked on projects involving early disease prediction and model evaluation, demonstrating the ability to handle structured data and compare model performance effectively.

Motivated to contribute to data-driven decision making and continuously improve technical and analytical skills

Technical Skills

- Predictive Modeling (Classification & Regression)
- Data Mining & Pattern Recognition
- Dimensionality Reduction (PCA)
- Data Cleaning & Preprocessing
- Interactive Data Visualization
- Model Evaluation & Performance Metrics
- Statistical Data Analysis

Tools & Technologies

- **Programming:** Python
- **Libraries:** Pandas, NumPy, Scikit-Learn, Matplotlib, Plotly, Seaborn
- **Platforms:** Google Colab, RapidMiner
- **BI & Dashboards:** Power BI, **Tableau**, Streamlit
- **Algorithms:** XGBoost, Random Forest, Logistic Regression, Linear Regression, Decision Trees

Project 1

Project Title

Job Market Analysis Dashboard

Description

This project focuses on analyzing job market data using an interactive dashboard to identify trends, demand, and key insights in the job market.

What I Did

- Cleaned and prepared job market dataset
- Built an interactive dashboard
- Analyzed job trends over time
- Visualized job distribution by city, company, and job titles
- Created charts for gender distribution and job demand

Tools

Power BI

Google Colab

Key Insights

- Total jobs analyzed: 1470
- Job demand varies across cities and companies
- Certain job titles appear more frequently (high demand)
- Gender distribution insights identified

Business Impact

- Helps understand job market demand
- Supports better career and hiring decisions

- Identifies high-demand roles and locations

https://app.powerbi.com/groups/me/reports/df98fb3b-81ef-4585-86cd-ef9dc4c8feab/dd2975ce44a037064ae8?experience=power-bi

https://colab.research.google.com/drive/1ikOcPEcAvqnwT5ox5dQ2FAYSKqBGOzSW?usp=sharing

Project 2

Project Title
Sales & Profit Performance Dashboard

Description
Developed a comprehensive visual analytics dashboard using Tableau to monitor and analyze business performance. The project focuses on tracking key performance indicators (KPIs) such as total sales, profit margins, and quantity sold across different segments and regions.

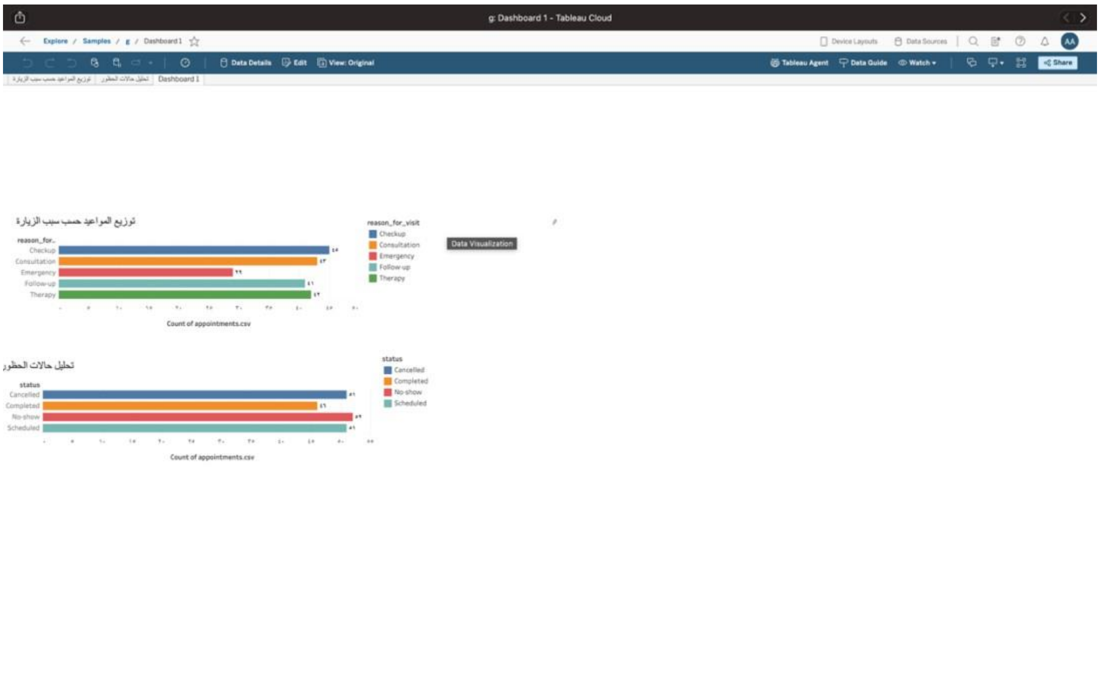
What I Did
- Connected and prepared sales datasets for visual analysis.
- Designed a central KPI overview to track Sales, Profit, and Quantity at a glance.
- Created a "Profit by Month" time-series analysis to identify seasonal trends and growth patterns.
- Developed a geographic map visualization to analyze sales distribution across various regions.
- Implemented segment-based analysis (Consumer, Corporate, Home Office) to identify the most profitable customer bases.
- Built a "Sales by Category" breakdown to compare performance across different product lines.

Tools Tableau, Data Visualization, Business Intelligence (BI).

Key Results
- Identified peak sales periods and months with the highest profit margins through time-series charts.
- Pinpointed high-performing geographical regions and underperforming areas for strategic focus.
- Visualized the correlation between sales volume and actual profit across different product segments.

Business / Real-World Impact
- Enables stakeholders to make data-driven decisions by providing a 360-degree view of business health.
- Optimizes resource allocation by identifying which segments and regions drive the most revenue.
- Simplifies complex sales data into actionable insights through an interactive and easy-to-navigate interface.

https://prod-in-a.online.tableau.com/t/471200340-00037986e5/views/g/Dashboard1


g: Dashboard 1 - Tableau Cloud
Explore / Samples / g / Dashboard 1
Device Layouts
Data Sources
AA
Data Details
Edit
View: Original
Tableau Agent
Data Guide
Watch
Share
توزيع المواعيد حسب سبب الزيارة
تحليل حالات الحظور
Dashboard 1
توزيع المواعيد حسب سبب الزيارة
reason_for_visit
Checkup
Consultation
Emergency
Follow-up
Therapy
Data Visualization
reason_for..
Count of appointments.csv
تحليل حالات الحظور
status
Cancelled
Completed
No-show
Scheduled
Count of appointments.csv

Project 3

Predictive Modeling in Real Estate and E-Commerce

Description

This project applies data mining techniques using regression and classification models to solve real-world problems in real estate and e-commerce.

What I Did

- Cleaned and prepared datasets
- Built a Linear Regression model for house price prediction
- Built a Decision Tree model for customer behavior prediction
- Split data into training and testing sets (70/30)
- Evaluated models using RMSE and Accuracy

Tools

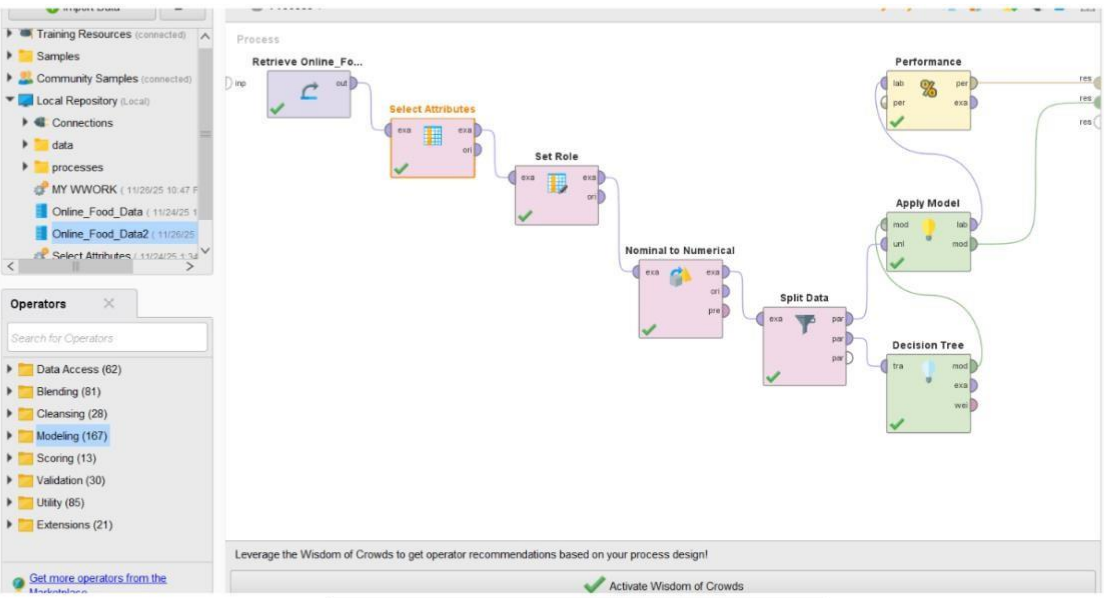
- RapidMiner, Machine Learning, Data Mining

Key Results

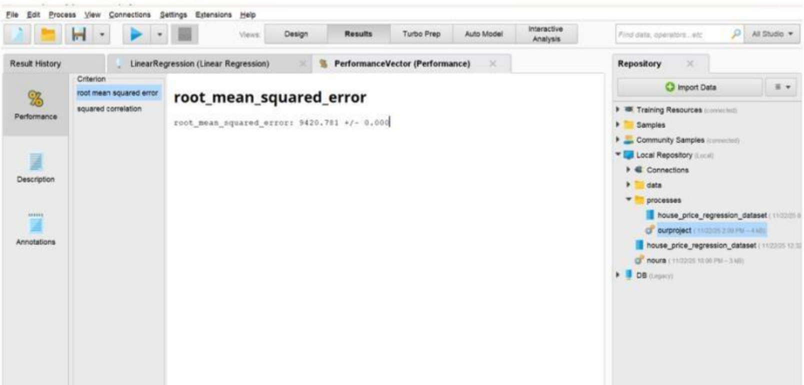
- RMSE ≈ 9762
- Accuracy = 83.82%
- Key features: Square Footage & Customer Feedback

Business Impact

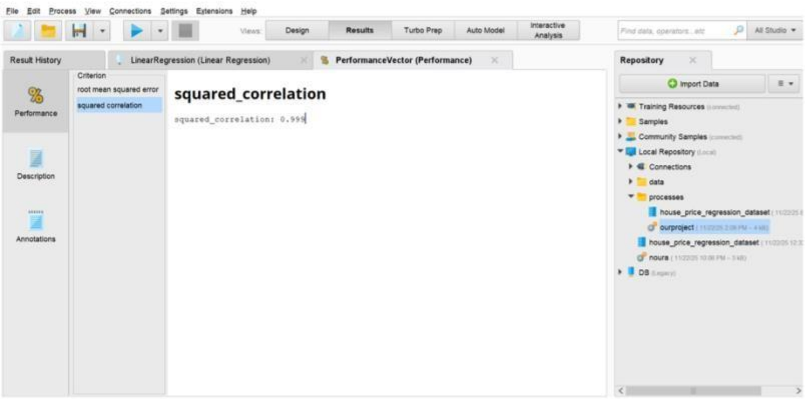
- Helps improve house price prediction
- Supports better customer retention strategies

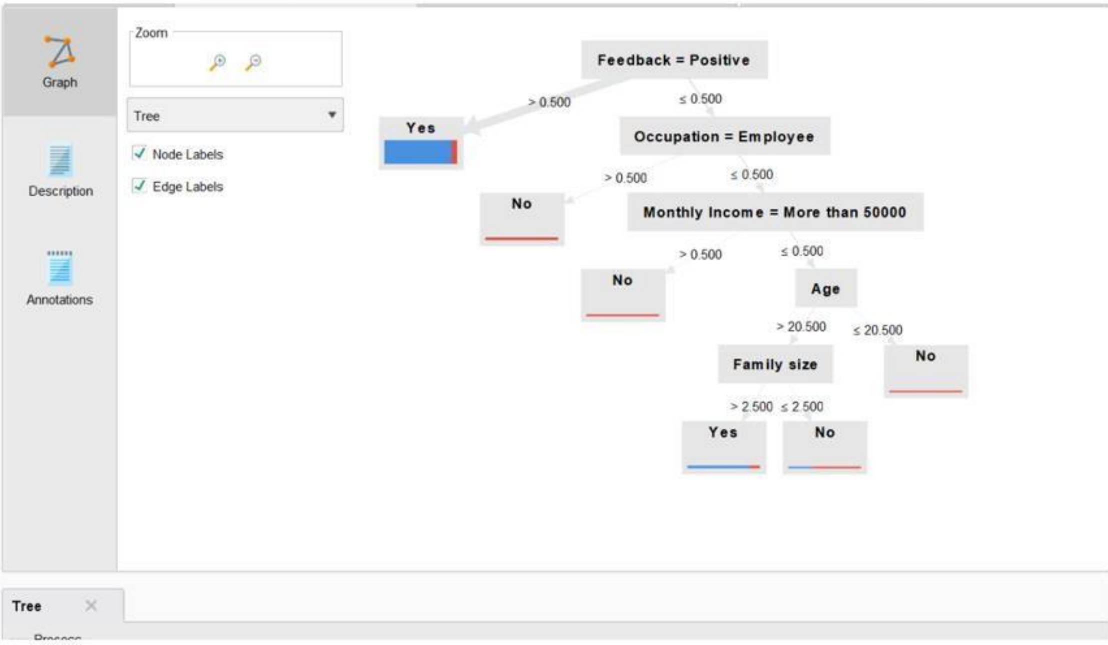


root_mean_squared_error (RMSE):



squared correlation (R^2):





Project 4

Project Title

Predicting Liver Disease Progression using Machine Learning

Description

This project focuses on predicting liver disease progression in patients with Hepatitis C using supervised machine learning techniques.

The goal was to classify patients into stable or progressive cases based on clinical and laboratory data to support early medical decision-making.

What I Did

- Preprocessed clinical dataset (handled missing values & encoding)
- Selected relevant medical features (lab tests, demographics)
- Built multiple ML models:
  - Logistic Regression
  - Random Forest
  - XGBoost
- Defined target variable (Stable vs Progressive)
- Split data into training/testing (80/20)
- Evaluated models using Accuracy, Precision, Recall, F1-score
- Compared model performance

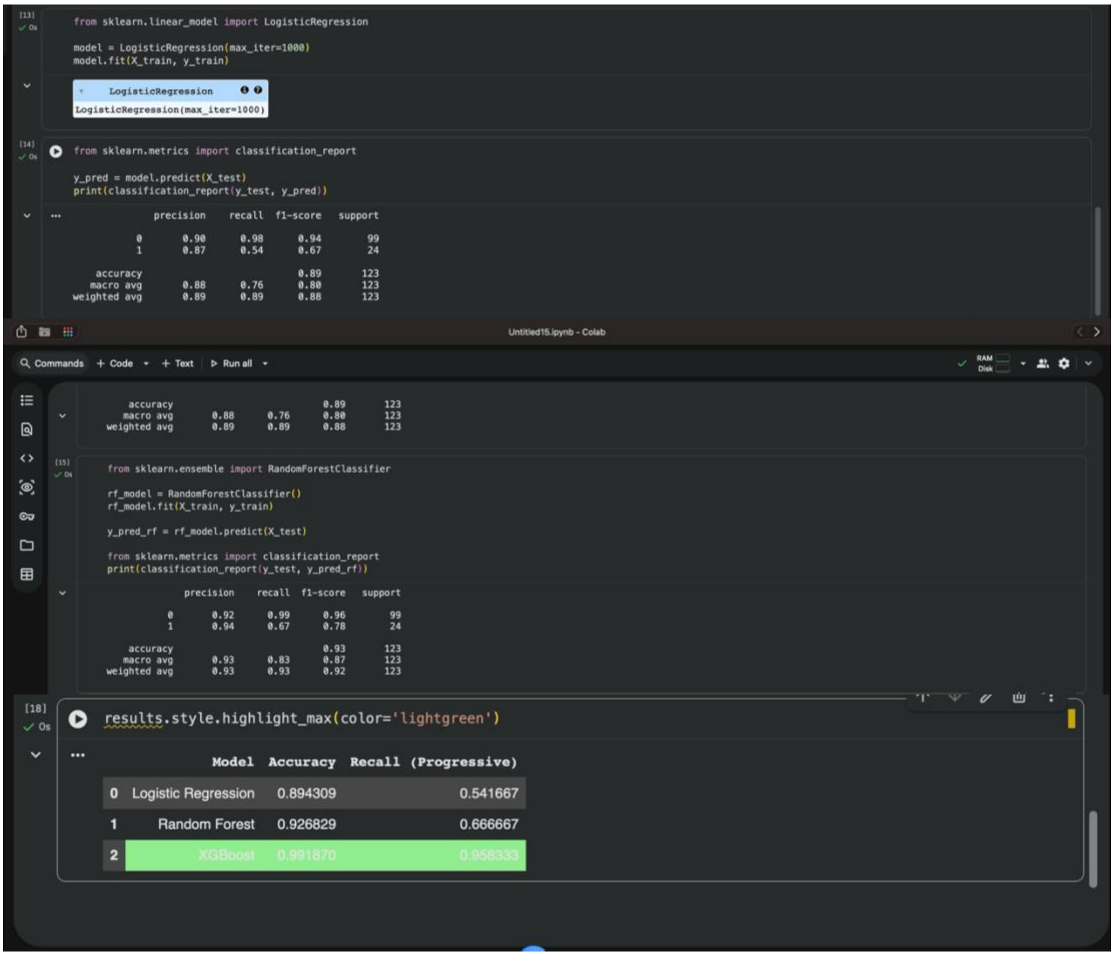
Tools

Python, Machine Learning, Data Preprocessing

Key Results

- XGBoost achieved 99% accuracy
- Random Forest achieved 93% accuracy
- Logistic Regression achieved 89% accuracy
- XGBoost showed best performance in detecting progressive case
- **Business / Real-World Impact**

- Enables early detection of disease progression
- Supports clinical decision-making
- Helps prioritize high-risk patients



```
[13]
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
[14]
from sklearn.metrics import classification_report

y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.90      0.98      0.94        99
           1       0.87      0.54      0.67        24

    accuracy                           0.89       123
   macro avg       0.88      0.76      0.80       123
weighted avg       0.89      0.89      0.88       123
```



```
    accuracy                           0.89       123
   macro avg       0.88      0.76      0.80       123
weighted avg       0.89      0.89      0.88       123
```


```
[15]
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier()
rf_model.fit(X_train, y_train)

y_pred_rf = rf_model.predict(X_test)

from sklearn.metrics import classification_report
print(classification_report(y_test, y_pred_rf))
```

```
              precision    recall  f1-score   support

           0       0.92      0.99      0.96        99
           1       0.94      0.67      0.78        24

    accuracy                           0.93       123
   macro avg       0.93      0.83      0.87       123
weighted avg       0.93      0.93      0.92       123
```

```
[18]
results.style.highlight_max(color='lightgreen')
```

	Model	Accuracy	Recall (Progressive)
0	Logistic Regression	0.894309	0.541667
1	Random Forest	0.926829	0.666667
2	XGBoost	0.991870	0.958333

https://colab.research.google.com/drive/1ikOcPEcAvqnwT5ox5dQ2FAYSKqBGOzSW#revisionId=0ByvndvSdy8i0TzgxcHFBUC94L1dXTExUQWdrU2dycHNkcU4wPQ&scrollTo=-JAyUBqFxF8Q

Project 5

Project Title

Dimensionality Reduction using Principal Component Analysis (PCA)

Description

- This project demonstrates the application of Principal Component Analysis (PCA) to reduce the dimensionality of the Iris dataset. The goal was to transform complex, multi-dimensional floral data into a 2D visualization while retaining as much variance as possible for easier analysis and classification.

What I Did

- • Loaded and explored the Iris dataset using Pandas to understand feature distributions.
- • Performed Data Preprocessing by separating features from labels and applying standard scaling to normalize the data.
- • Implemented PCA to reduce the dataset from four dimensions to two principal components.
- • Built a new dataframe containing the principal components and original targets for visualization.
- • Created a 2D scatter plot using Matplotlib to visualize the distinct clusters of Iris species (Setosa, Versicolor, and Virginica).

Tools

- Google Colab, Python, Scikit-Learn (PCA, StandardScaler), Pandas, Matplotlib.

Key Results

- • Successfully reduced 4 features (sepal/petal length and width) into 2 principal components.
- • The 2D visualization clearly separated the 'Setosa' species from 'Versicolor' and 'Virginica', confirming the effectiveness of PCA in capturing variance.
- • Maintained data integrity after scaling, ensuring accurate principal component calculation.

Business / Real-World Impact

- • Simplifies high-dimensional datasets for easier human interpretation and visualization.
- • Improves machine learning model efficiency by reducing the number of input variables without significant information loss.
- • Assists in identifying patterns and clusters in biological or technical data that are not visible in higher dimensions.

Project 6

Project Title
Interactive Dimensionality Reduction & Visualization

Description
This project focuses on applying Principal Component Analysis (PCA) to the Iris dataset and visualizing the results through interactive plots. The objective was to transform high-dimensional data into a lower-dimensional space while providing an interactive interface for deeper data exploration.

What I Did

- Imported and prepared the Iris dataset within a Google Colab environment.
- Standardized the features using StandardScaler to ensure all variables contribute equally to the analysis.
- Applied PCA to reduce the dataset dimensions to 2 principal components.
- Developed an interactive scatter plot using Plotly to represent the transformed data points.
- Mapped original species labels (Setosa, Versicolor, Virginica) to the visual components for clear classification.

Tools
Google Colab, Python, Scikit-Learn (PCA, StandardScaler), Pandas, Plotly.

Key Results

- Successful reduction of the feature space while maintaining the structural integrity of the clusters.
- Created a dynamic visualization where different species are clearly separated and identifiable.
- Achieved a functional workflow for scaling and transforming data efficiently.

Business / Real-World Impact

- Enhances data storytelling by allowing users to interact with and explore complex data patterns.
- Reduces computational overhead for subsequent machine learning tasks by simplifying input features.
- Provides a scalable template for visualizing high-dimensional datasets in various industries.

https://colab.research.google.com/drive/1ikOcPEcAvqnwT5ox5dQ2FAYSKqBGOzSW#revisionId=0ByvndvSdy8i0TzgxcHFBUC94L1dXTExUQWdrU2dycHNkcU4wPQ&scrollTo=-JAyUBqFxF8Q